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RESEARCH ARTICLE

Understanding the relationship between spatial urbanization and urban heat island using machine learning: a landscape-based spatial framework

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Abstract

The urban heat island (UHI) effect driven by urbanization induced land use and land cover (LULC) change has emerged as one of the serious environmental issues. Urbanization is one of the prime drivers of the UHI effect in cities. Therefore, it is essential to understand the impact of the urbanization on UHI. This study aims to understand the relationship between urbanization and surface urban heat island (SUHI) using machine learning approach in Kolkata Metropolitan Area (KMA), India. A spatial urbanization index (SUI) has been developed to assess the spatial urbanization and its relationship with UHI using spatial landscapes metrics. A spatial gradient approach was also applied to understand the changes of land surface temperature (LST) along urban–rural gradient (URG). A correlation and regression analysis were employed to understand the relationship between urbanization and SUHI. The findings showed that the impervious surface (IS) substantially increased from 2000 to 2020 with a net increase of 128%. The area under high and very high SUHI intensity increased by 287%, and 207% The IS and LST from the core decreased with increasing distance along URG. The RF model showed that the area mean (AREA_AM) (built-up) had the greatest influence on spatial urbanization, followed by class area (CA), Cohesion (COHE), and percentage of landscape (PLAND). Among the spatial landscape metrics, largest patch index (LPI) had the greatest influence on SUHI followed by patch density (PD), division (DIV), number of patches (NP), and Interspersion and Juxtaposition index (IJI). Lastly, the spatial urbanization had a positive correlation with SUHI intensity. The findings emphasize the pressing necessity of sustainable urban planning strategies to alleviate the detrimental consequences of urban expansion on thermal environments. The integration of green and blue infrastructure into urban planning policies is essential for the effective combat of UHI.

Keywords Kolkata metropolitan area · Urban heat island · Impervious surface · Land surface temperature · Urban management

Introduction

The accelerated urbanization, inadequate infrastructure, and socio-economic disparities in Global South cities present distinctive challenges resulting from urban heat island (UHI) effects (Ramakreshnan et al. 2019; Li et al. 2024). Unplanned urban expansion is a common occurrence in numerous cities in the Global South, resulting in the loss of green spaces, expansion of impervious surfaces, and inadequate land use planning (Miah et al. 2024). These factors exacerbate heat stress, particularly for vulnerable populations, by increasing urban temperatures (Lin et al. 2023; Arshad et al. 2020). The effects of UHI are further exacerbated by the limited access to natural infrastructure, such as urban forests and street trees (Marando et al. 2022;

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Mirabi and Davies 2022). The potential for natural cooling mechanisms is diminished by insufficient vegetation cover and fragmented vegetation (Ferrini et al. 2020). Furthermore, the prevalence of low-income settlements with high population densities and inadequate ventilation results in an increase in indoor heat exposure, which in turn contributes to severe health risks, such as heat-related illness and mortality (Laue et al. 2022). Another critical issue is energy poverty. Numerous urban residents are compelled to depend on inefficient methods that offer minimal respite due to their lack of access to affordable cooling solutions, such as air conditioning. For example, the people living in slums are highly vulnerable due to extreme heat conditions and heat waves (Rashid et al. 2013; Mukhopadhyay and Weitz 2022). In the cities of developing countries, increase of temperature and frequent occurrence of heat waves have been a major environmental issue (Arshad et al. 2020). The rapid urbanization and urbanization induced transformation of the land use and land cover (LULC) is highly responsible to deteriorate thermal environmental conditions in cities. Therefore, it is very important to understand the relationship between urbanization and UHI in cities for climate mitigations.

The UHI is closely linked to the spatial dynamics of urbanization, where increased human activities and infrastructural development lead to elevated temperatures in urban areas compared to their rural counterparts (Zhang et al. 2013; Kong et al. 2021). The UHI effect is primarily driven by alterations in land cover, characterized by a transition from vegetation to impervious surfaces which absorb and retain heat (Irfeeyet al. 2023). The previous studies showed that urbanization significantly modifies local climates by reducing vegetation cover and generating heat from human activities (Patel et al. 2021). A study by Li et al. (2022) demonstrated that urban expansion is directly correlated with UHI intensity, where denser urban areas exhibited higher temperature anomalies. This correlation is further supported by studies using remote sensing, revealing that areas with extensive green spaces experience less severe UHI effects (Chen et al. 2023). In this study, surface urban heat island (SUHI) has been developed using land surface temperature (LST). The intensity and impacts of SUHI are significantly influenced by socio-economic factors. Poor housing conditions, restricted access to cooling infrastructure, and diminished adaptive capacity render low-income communities particularly susceptible to UHI (Yao et al. 2020a, b). A significant number of low-income communities are more susceptible to extreme heat stress due to their higher population densities, inadequate ventilation, and a lack of green spaces. Air conditioning, heat-resistant building materials, and homes in areas with improved environmental planning are accessible to wealthier residents, while lower-income groups reside in heat-prone areas with inadequately insulated structures (Wang and Zhou 2022). In many developing cities, informal settlements are prevalent, but their lack of appropriate urban planning results in increased exposure to high temperatures (Grover and Singh 2016; Li et al. 2024). Additionally, employment conditions are a factor, as outdoor workers, including street vendors and labours, are subjected to protracted heat exposure, which can result in health complications such as dehydration and heatstroke. Furthermore, the unequal distribution of green infrastructure (GI) is a consequence of disparities in urban planning. Wealthier regions are able to take advantage of tree-lined avenues and parks that aid in the mitigation of UHI, while impoverished neighbourhoods remain susceptible. Equitable urban policies, investment in heat-resilient infrastructure, expansion of urban green spaces, and community-based adaptation strategies to safeguard vulnerable populations are all necessary to address the socio-economic dimension of UHI. Urban rural gradient (URG) approach has been used to understand the changes in LST from the centre of the city towards the outside. Studies showed that impervious surface coverage correlates with increased LST, especially in densely urbanized zones, while vegetated rural areas exhibited cooling effects due to higher evapotranspiration and shading (Chen et al. 2022). Moreover, URG analysis helps to identify thresholds at which impervious surface (IS) significantly affects UHI, offering insights into optimal urban planning (Wang and Zhou 2022; Almeida et al. 2021). It also emphasizes the importance of integrating green infrastructure to mitigate LST in peri-urban areas (Huang et al. 2023). This approach provides a multiscale perspective for developing sustainable urban designs that balance development with ecological processes, particularly in mitigating UHI effects and reducing energy consumption (Sun et al. 2021).

A study by Dutta et al. (2022) emphasized that the reduction of green spaces and water bodies, crucial for natural cooling, has heightened UHI intensity. Additionally, Roy et al. (2022) found that the densification of the city and the sprawling urban areas contributed to rising temperatures, with the core regions showing the most prominent UHI effects. Urban growth has outpaced the implementation of nature-based solutions (NBS) like green infrastructure and street trees, which are known to mitigate UHI. Furthermore, high population density in Kolkata may exacerbate heat stress, making vulnerable communities more susceptible to heat-related risks. Overall, urbanization may lead UHI problem worse, in Kolkata necessitating sustainable urban planning and the promotion of green spaces. This finding of the previous research inspired the researcher to carry out a study to understand the relationship between urbanization and UHI in Kolkata Metropolitan Area (KMA), India. Thus, the key objective of this study is to understand the relation between urbanization and UHI in KMA. The specific objectives of the study are to a) understand the spatio-temporal dynamics of IS from 2000 to 2020 in KMA, b) assess the spatio-temporal

dynamics of BCI, LST and SUHI in KMA from 2000 to 2020 and lastly c) evaluate the relative importance of the spatial metrics affecting SUHI and urbanization, respectively. Thus, three basic research questions have been formulated based on the objectives. These were a) what are the spatial and temporal changes in IS in KMA from 2000 to 2020? b) what are the spatio-temporal patterns of the intensity of BCI, LST and SUHI in KMA? and c) In KMA, which landscape metrics have the most substantial impact on the intensity of SUHI and urbanization?

Material and methods

Study area

KMA physio-graphically is located in lower Gangetic plain. KMA is located along the bank of river Hoogly (Fig. 1). The most important urban centers in KMA are, Kolkata, Chandannagar, Howrah, Kalyani. Kolkata is one of the important trade and commercial center in Eastern India. It comprises total 14.06 million population with a density of 7480 persons/km^2 (Chatterjee and Dinda 2022). Kolkata and its surrounding areas are growing rapidly with an annual growth of 2% over last few decades (Mukherjee 2012). The rapid increase of built-up areas will significantly affect thermal environment in KMA.

UHI vulnerability in Kolkata has increased significantly due to rapid urbanization. Studies highlighted that rapid expansion in Kolkata, coupled with unplanned development, has intensified UHI effects, exacerbating surface and atmospheric temperatures across the city (Mandal et al. 2022; Chatterjee and Majumdar 2022). The main contributors to UHI in Kolkata include reduced vegetation, increased built-up areas, and the extensive use of impervious materials like concrete and asphalt (Halder et al. 2021). These changes have led to the absorption and retention of heat, especially during the summer months, thereby creating localized hotspots.

Research framework

Data source and processing

Landsat images extracted from the United States Geological Survey (USGS) were used to develop built-up expansion, biophysical composition index (BCI), LST, SUHI for the year 2000, 2010 and 2020. Two 5 thematic mapper (TM) (2000 and 2010) and one Landsat 8 operational Landsat imager (OLI) (2020) were used for built-up expansion, BCI, LST, and SUHI.

The built-up expansion maps were developed to develop IS from LULC in KMA using maximum likelihood supervised classification in 2000, 2010 and 2020. A total of six classes were identified, these are built-up, vegetation cover, water bodies, agricultural land and bare lands. Except built-up, all the five classes were excluded to show the pattern of built-up expansion in the last 20 years in KMA.

The temperature from the core of the city towards the outside decreases with increasing distance. This temperature gradient is known as urban–rural gradient (URG). As urbanization intensifies, IS such as roads and buildings increase, disrupting the natural energy balance and elevating LST, thereby contributing to UHI effect (Li et al. 2020). By assessing spatial patterns across urban to rural gradients, this analysis reveals how LST varies with changes in land cover composition (Zhao et al. 2021). The URG analysis was carried out to understand the pattern of IS and the impact of IS on the UHI. The multiple-ring buffer zones were created from the centers with an interval of 1 km covering up to 15 km. The IS (%) and average LST were extracted from each buffer zone during the entire study period (2000 to 2020). The proportion of IS area was calculated to investigate the changes of IS along the URG. The BCI proposed by Deng and Wu (2012) was used to illustrate IS abundance and develop IS maps from 2000 to 2020. BCI is a composite index used to construct an urban environmental index to distinguish terrestrial materials in an urban environment. It is based on the V-I-S model (vegetation-impervious-soil model) and Tasseled Cap (CP) transformations. BCI was widely used in previous studies to develop IS maps due to their effectiveness in delineating IS and various spatial resolution images (Meng et al. 2018; Yao et al. 2020a). The following equation was used to develop BCI [1]:

$$BCI = \frac{(H+L)/2 - V}{(H+L)/2 + V} \tag{1}$$

where H [2], L [3], and V [4] refer to thehigh albedo (TC1), vegetation (TC2) and low albedo (TC3). The following equation was used to calculate the factors:

$$H = \frac{TC1 - TC1_{\min}}{TC1_{\max} - TC1_{\min}} \tag{2}$$

$$V = \frac{TC2 - TC2_{\min}}{TC2_{\max} - TC2_{\min}} \tag{3}$$

$$L = \frac{TC3 - TC3_{\min}}{TC3_{\max} - TC3_{\min}} \tag{4}$$

where TCi refers to the first three TC components and TCimin and TCimax refer to the minimum and maximum value of ith components.

Retrieval of LST

LST maps were developed for the years 2000, 2010, and 2020 using Landsat TM (for the years 2000, and 2010) and Landsat 8 OLI (2020), respectively. Two Landsat 5 TM and two Landsat OLI 8 images downloaded from USGS were used to develop LST maps. The extraction of LST is essential for better monitoring of the urban thermal environment (Chen et al. 2020). The image-based method was applied to extract LST from the thermal bands (band 6 for Landsat 5 TM and band 10 for Landsat 8 OLI). The following steps were used to extract LST from the thermal bands: Firstly, the digital number (DN) has been converted into spectral radiance [5]:

$$L_\lambda = M_L \times Q_{CAL} + A_L \tag{5}$$

where L refers to the spectral radiance (W/(m2.sr. µm)), ML refers to the radiance multiplicative scaling factor (RADIENCE MULTI BAND_X), AL refers to the radiance additive scale (RADIENCE_ADD_BAND_X) where X refers to the number of band, and QCAL refers to the calibrated DN pixel value.

Secondly, spectral radiance has been converted at satellite brightness temperature using the following equation [6]:

$$T_B = \frac{K_2}{ln\left(\frac{K_1}{L\lambda} + 1\right)} \tag{6}$$

Thirdly, the at-brightness temperature was converted into emissivity. The following equation was used to calculate the final LST. K_1 and K_2 is 607.76 and 1260.56 for Landsat 5 TM and 774.8853 and 1321.0789 for Landsat 8 OLI [7].

$$T_S = \frac{T_B}{1 + \left(\lambda \times T_B/\rho\right)\ln(\varepsilon)} \tag{7}$$

where λ refers to the central wavelength of the thermal band (11.45 µm for Landsat 5 TM, and 10.895 µm for Landsat 8 OLI), p= hXc/$^{\delta}$ (1.438X̊10-2mk), $^{\delta}$ refers to the Boltzmann constant (1.38X10-23/J/K), h= Planck's constant

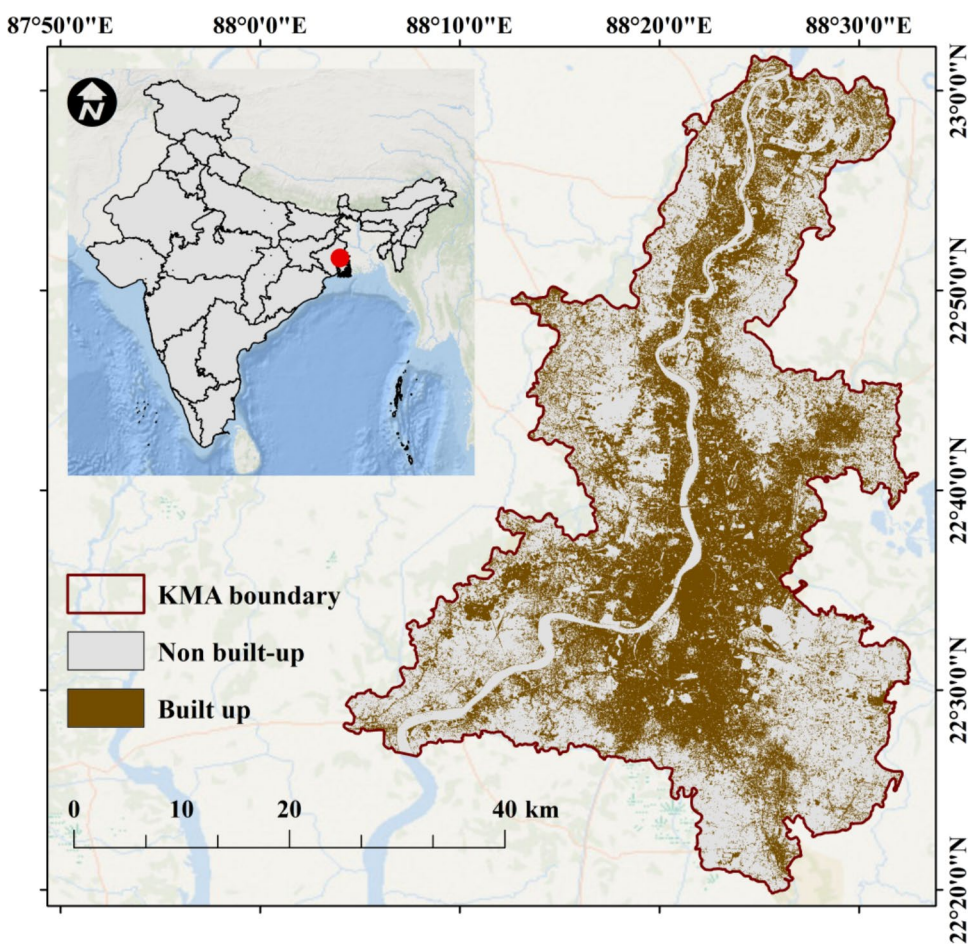

Fig. 1 Location map of the study area. Kolkata Metropolitan Area (KMA)

(6.626X10-34Js), c refers to the velocity of the light (2.998X108 m/s).

Development of SUHI

The cities of the world frequently experience extreme heat conditions due to rise in temperature leading the public health vulnerability. Therefore, it is essential to assess and map UHI to mitigate heat in cities. UHI can be assessed from two perspectives i.e. based on the air temperature known as AUHI and surface temperature known as SUHI. In this study, a SUHI was developed to understand the relationship between SUHI and urbanization in KMA from 2000 to 2020. The SUHI has been assessed based on the LST extracted from thermal band (band 10).

In previous studies, the UHI index was widely used to estimate the pattern of the SUHI effect (Mathew et al. 2017). The following equation was used to construct SUHI intensity [8]:

$$SUHI_{index} = \frac{LST_i - LST_{min}}{LST_{max} - LST_{min}} \tag{8}$$

where LST_i refers to the LST of a specific pixel in the entire study area, LST_{max} and LST_{min} refer to the maximum and minimum value of LST in entire study area.

Development of spatial urbanization index (SUI)

The urbanization has four broad perspectives, these are: population urbanization, social urbanization, economic urbanization and spatial urbanization or land urbanization (Bai et al. 2018). The urban land expansion is the spatial manifestation of urbanization (Peng et al. 2017). In this assessment of urbanization effectiveness, spatial urbanization has been measured due to lack in obtaining economic (e.g., gross domestic production-GDP), social (e.g., energy consumption) and population data, spatial urbanization has been measured based on the spatial metrics. In this study, an urbanization index i.e., SUI has been developed to understand the relationship between SUI and SUHI. A spatial urbanization (i.e. land urbanization) approach has been adopted to develop SUI. Spatial urbanization refers to the expansion of the land, such as proportion of construction land, per capita construction land. The landscape metrics were used to develop spatial or land urbanization in this study. A total of fifteen spatial metrics has been selected to develop a comprehensive SUI (Table 1). The following equation has been employed to calculate SUI:

$$SUI = \frac{SU_{ij} - SU_{i_{min}}}{SU_{i_{max}} - SU_{i_{min}}} \tag{9}$$

where, SU_{ij} refers to the standardized value of observed parameters, $SU_{i_{min}}$ refers to the minimum value among the indices, and $SU_{i_{max}}$ refers to the maximum value among the indices.

A spearman correlation (significant at < 0.05) was carried out to find out the relationship between LST and IS for each year and LST and landscape effectiveness factors. A linear regression was also performed to assess the impact of spatial metrics (independent factors) on spatial urbanization (dependent factors). All the spatial mapping were developed in ArcGIS environment (version 10.3) and statistical analysis was performed using statistical packages for social science (SPSS version 26).

Factors affecting urbanization and SUHI

Random forest (RF) method has been applied to understand the relative importance of spatial urbanization factors and impact of spatial urbanization factors on UHI. The RF machine learning method plays a pivotal role in understanding the relationship between spatial urbanization and UHI effects due to its robust predictive capabilities and ability to handle complex, non-linear data interactions. RF is highly effective in modeling spatial heterogeneity and capturing the diverse environmental variables influencing UHI, such as LULC changes, vegetation, and IS (Zhao et al. 2020). Unlike traditional regression models, RF can assess the relative importance of multiple predictor variables, allowing researchers to pinpoint key drivers of UHI, such as increased built-up areas and reduced green cover (Li et al. 2021). Thus, RF's capacity to handle high-dimensional data and its interpretability make it an indispensable tool for exploring the complexities of urbanization and its impacts on UHI, supporting efforts toward climate-resilient urban planning. In this study, RF regression model has been used to identify the relative importance of the spatial urbanization factors affecting UHI.

Results

IS subsequently increased from 2000 to 2020 in KMA (Fig. 2). The areas with high IS were highly concentrated along the river (Hoogly river). The higher IS was found between 3 to 5 km away from the center of the city, indicating the fact that IS area was increased by infilling and expansion of urban areas. During the period 2000 to 2010, the IS area was higher at the distance of 4 and 5 km, which indicated that IS areas expanded by infilling, expansion and leapfrog. Between 2010 to 2020, IS areas were mainly expanded by leapfrog and expansion at the distance of 8 km to 15 km (Fig. 3).

The results showed that the BCI value was higher in built-up areas, and the land cover types with vegetation and water bodies showed a lower BCI (Fig. 4).

There was a dramatic increase and spatial variation of LST across KMA. The mean LST in 2000 was 27.71^{O}C, which increased to 35.58^{O}C in 2020 (mean LST increased by about 28.40% from 2000 to 2020). The maximum and minimum LST in 2000 was 36.83^{O}C and 24.11^{O}C, reaching 48.94^{O}C and 27.57^{O}C in 2020. The core areas of Kolkata were characterized by relatively higher LST in comparison to fringe areas. From 2000 to 2020, maximum LST was increased by about 32.88%, with the highest from 2000 to 2020 (Fig. 5). There was also a substantial increase in areas with high and very high SUHI from 2000 to 2020. For example, in 2000, only 0.90% and 0.055% of the total area was under high and very high SUHI, which increased to 4.39% and 0.169% in 2020. The areas under high and very high increased by about 287% and 207% in the last 20 years (from 2000 to 2020). On the other hand, the areas with very low and low SUHI decreased by about 28.55% and 35.10% from 2000 to 2020. The areas under very low and low were 10.43% and 80.02%, which declined to 7.45% and 51.90%, respectively. Thus, the areas with high and very SUHI significantly increased from 2000 to 2020 in KMA (Fig. 6).

The regression analysis showed the relationship between LST and IS from 2000 to 2020 (Fig. 7). In 2000, a strong correlation ($R^2 = 0.982$) indicates that a 1% increase in IS raises LST by 26.10 °C. By 2010, the impact decreases slightly ($R^2 = 0.897$) with a 20.64 °C rise per 1% IS increase. In 2020, the correlation weakens further ($R^2 = 0.793$), with LST increasing by 9.324 °C per 1% IS. Over time, the reduced impact suggests changes in urban infrastructure or other factors moderating the UHI effect, lessening the link between IS and LST.

There was positive correlation between SUHI and SUI indicating the fact that SUHI intensity increases with increasing spatial urbanization (Fig. 8).

The relative importance of spatial metrics on the spatial urbanization and UHI has been assessed using RF regression. The R^2 value of the proposed model was 0.87 implying 87% of the variance in spatial urbanization was explained by fifteen spatial metrics (Table 2). AREA_AM had the greatest influence on spatial urbanization, followed by CA, COHESION, PLAND, PARA_MN, CLUMPY, and LPI (Fig. 9). On the other hand, the factors affecting spatial urbanization had also significant impact on UHI. Among fourteen spatial metrics, LPI had the greatest influence on the UHI followed by PD, DIVISION, NP, and IJI. The R^2 value of the proposed model was 0.48 implying 48% of the variance in thermal environment was explained by fifteen spatial metrics (Table 3).

Discussion

The results showed that IS abundance showed a positive relationship with LST. It clearly denoted the fact that IS has a significant positive impact on LST in KMA. IS such as concrete and asphalt, increase LST by absorbing more solar radiation due to their low albedo (Lu et al. 2023; Ghenai et al. 2023). These materials store heat during the day and slowly release it at night, intensifying the UHI effect (Mohajerani et al. 2017). Unlike vegetation, IS lack evapotranspiration, reducing natural cooling. They also prevent moisture retention, eliminating evaporative cooling. Additionally, urban geometry traps heat, limiting airflow and further raising temperatures. The combination of high thermal conductivity, heat storage, and restricted cooling mechanisms makes IS a key driver of elevated LST in urban environments, exacerbating heat stress and climate challenges. This finding is similar to other previous studies (Wang et al. 2022; Xu et al. 2013). Wang et al. (2019) carried out a study in Babol City in Iran to find out the relationship between IS and LST and the findings showed that there was a positive relationship between IS and LST. Hua et al. (2020) performed a study in Xiamen City in Southeast China during the summer season from 1989 to 2016, and their findings showed that LST had a strong positive co-relationship with IS area. In addition to this, it was documented that the impact of IS areas on LST became more prominent with urbanization. The increase in LST with increasing IS area was mainly due to urban infrastructural development, intensive LULC (Zhang and Sun 2019). Morabito et al. (2021) carried out a study in Italian metropolitan cities to understand the role of IS and tree cover in the formation of SUHI. Their findings showed that LST increases with increasing the density of IS and decreasing tree cover density. The water bodies in urban areas significantly affect LST, followed by IS vegetation cover (Deilamiet al. 2018; Kong et al. 2014). Peng et al. (2018) and Zhou et al. (2011) stated that the role of water bodies in controlling LST depends on the area of the water bodies. Thus, it can be stated that water bodies have a cooling effect on the LST distribution (Gupta et al. 2019). The results of the study showed that mean LST subsequently decreased with increasing distance from the centers and decreasing IS areas. For example, in 2020, the mean LST was 37.34^{O}C at a 1 km buffer, which decreased to 35.34^{O}C at a 15 km buffer. It clearly denotes the fact that LST decreases along URG. In previous studies, it was well documented that there was a significant change in LST along the urban–rural gradient (Estoque et al. 2017).

It was observed that the previous findings showed similar results. For example, Qiao et al. (2018) carried out a

Table 1 Spatial metrics used to develop SUI. SUI represents spatial urbanization index

Spatial Metrics	Acronyms	Explanation in relation to UHI	Reference
Area-weighted mean patch area	AREA_AM	Indicates the average size of patches, weighted by area. Larger patch sizes can mitigate UHI by providing more green spaces that cool the environment	Li et al. 2020
Class area	CA	Represents the total area covered by a land-use type. Higher green or water class area can lead to more cooling effects, reducing UHI intensity	Wu et al. 2019
Cohesion index	COHESION	Measures the physical connectedness of patches. Higher cohesion of green patches can enhance cool air flow, mitigating UHI	Buyantuyev and Wu 2010
Percentage of landscape	PLAND	Refers to the proportion of the landscape composed of a particular land cover. A higher proportion of green areas generally help reduce UHI by lowering LST	Zhou et al. 2022
Perimeter-area ratio mean	PARA_MN	Indicates the complexity of patch shapes. Lower ratios suggest more compact patches, which may help reduce UHI by maintaining cooler microclimates	Li et al. 2021
Clumpiness index	CLUMPY	Quantifies the clustering of patches. Clumped green spaces can cool larger areas, helping in UHI mitigation	Wang et al. 2019
largest patch index	LPI	Measures the size of the largest patch within the landscape. Larger green or water patches tend to have more significant cooling effects, reducing UHI	Zhang et al. 2023
Interspersion juxtaposition index	IJI	Represents how patches are distributed and interspersed. Even distribution of green spaces enhances cooling throughout the city, reducing UHI	Jin et al. 2018
Aggregation index	AI	Refers to the degree to which patches of the same class are aggregated. Higher aggregation of green spaces is beneficial for reducing UHI	Lin et al. 2020
Patch density	PD	Describes the number of patches in the landscape. More patches, if vegetated, can contribute to mitigating UHI by creating microclimates	McGarigal and Marks (1995)
Mean patch area	AREA_MN	The average size of patches in a landscape. Larger patches, especially vegetated ones, contribute to cooling the surrounding area, mitigating UHI	Herold et al. 2003
Division Index	DIVISION	Measures the degree to which the landscape is fragmented. Higher fragmentation may increase UHI by reducing the effectiveness of green patches in cooling	Wu, 2013
Landscape shape index	LSI	Measures the complexity of patch shapes. More complex shapes may not be as effective in UHI mitigation as more compact patches	Oke and Cleugh (1987)
Shape mean index	SHAPE_MN	Represents the mean shape complexity of patches. Less complex shapes (closer to circles) are more effective in mitigating UHI due to lower edge effects	Li et al. (2018)
Number of patches	NP	Indicates the total number of patches in the landscape. A higher number of green patches may help to mitigate UHI, but overly fragmented patches may reduce effectiveness	McGarigal and Cushman (2002)

study in Beijing (China), and their findings showed that IS increased by about 144% from 1990 to 2015. Xu et al. (2018) performed a study in Guangzhou (China), and the results showed IS increased by about 725%, with an annual increase of 25.92%. Similar findings were reported by Li et al. (2019), Adeyemi et al. (2021), Zhang et al. (2021). In India, a number of research studies were carried out on the urban land use land cover dynamics, and the results of the study showed a dramatic increase in IS. For example, Bharath et al. (2018) carried out a study of the Indian cities Delhi, Pune, Mumbai, Chennai and Coimbatore, and their findings showed a significant increase in built-up areas over the periods of time. Jain et al. (2016) also carried in Delhi from 1977 to 2014, and their results showed that IS increased by about 400%. In Mumbai, a study was carried out by Rihan et al. (2021) from 1991 to 2018, and their results showed that IS was increased by about 99.90%. Thus, similar findings were reported from many other studies across the Indian cities (Kotharkar and Bagade 2018; Siddiqui et al. 2021). In this study also, it was observed that IS increased by 185% from 2000 to 2020. In

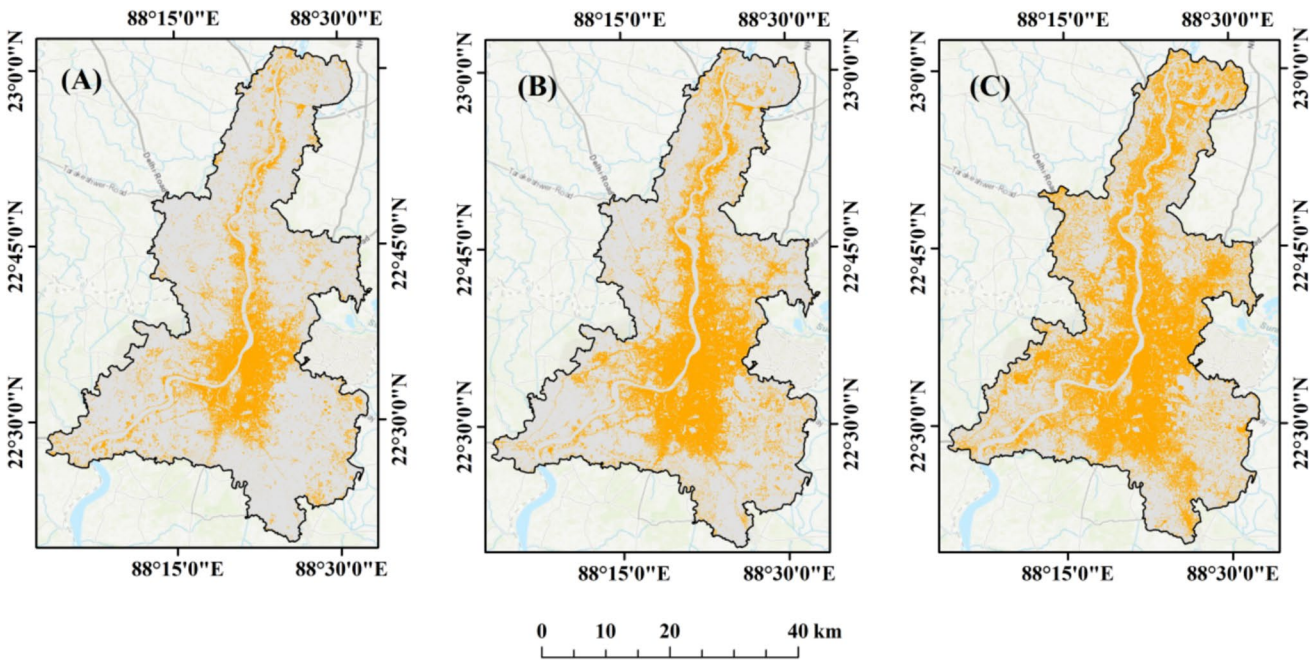

Fig. 2 Spatio-temporal pattern of built-up expansion (**A**) 2000, (**B**) 2010, and (**C**) 2020

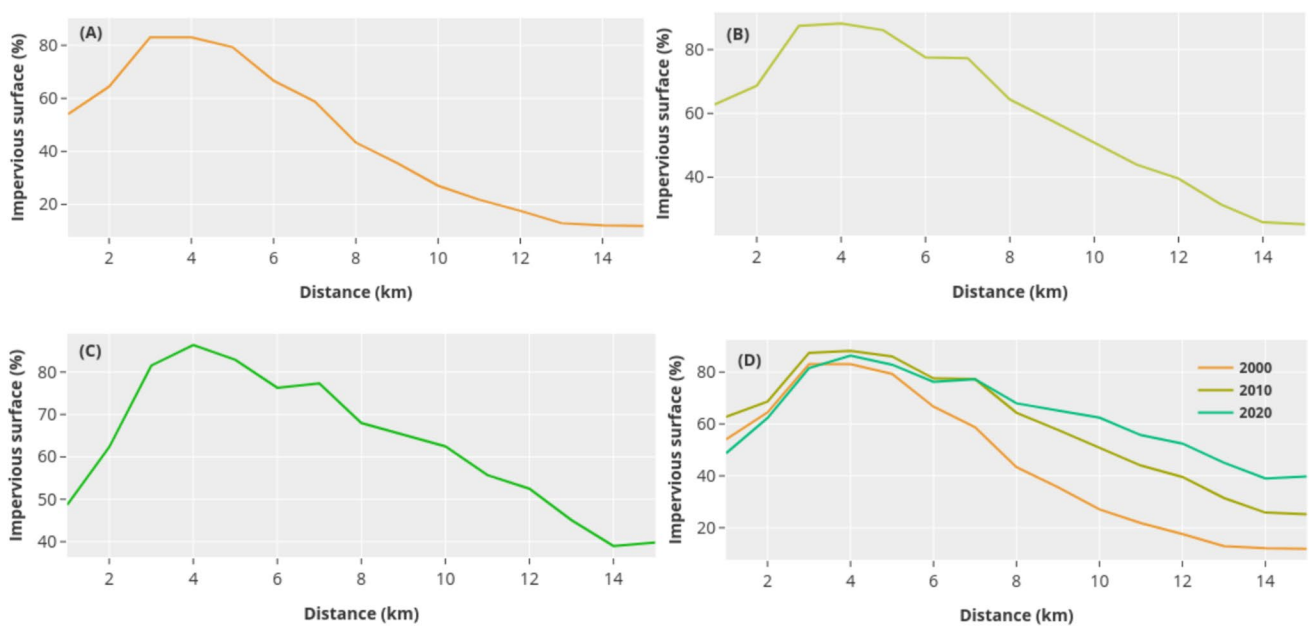

Fig. 3 Changes in IS across the buffer (**A**) 2000, (**B**) 2010, and (**C**) 2020 (**D**) 2000 to 2020. IS means impervious surface. IS refers to the impervious surface

2000, the IS area was 271 km^2, which increased to 776.40 km^2.

In previous literature, it was documented that the UHI intensity significantly increased., similar findings were reported. For example, Meng et al. (2018) and Yao et al. (2020b) carried out a study in Beijing (China), and their results showed that SUHI increased from 2003 to 2015 and 2009 to 2018. Similar results were observed in previous literature (Dewan et al. 2021; Rahman et al. 2022). In a number of research studies in India, it was well-recognized that

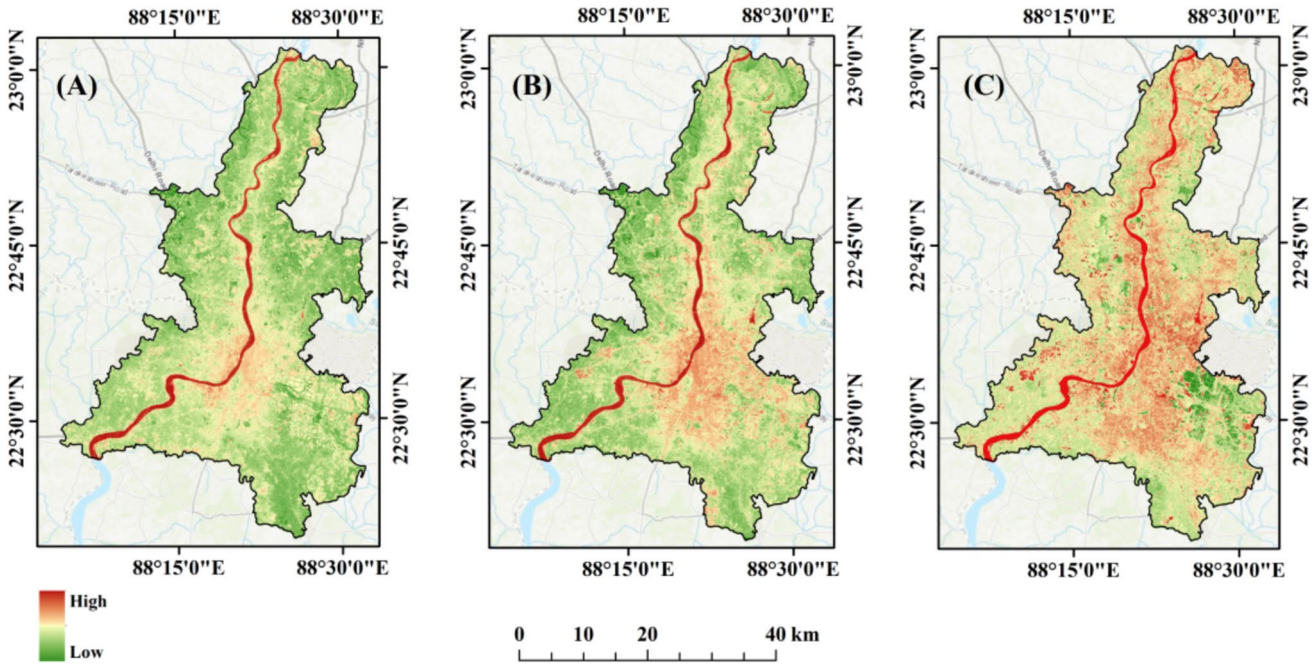

Fig. 4 Spatio-temporal pattern of BCI (**A**) 2000, (**B**) 2010, and (**C**) 2020. BCI means biophysical composition index. BCI refers to the biophysical composition index

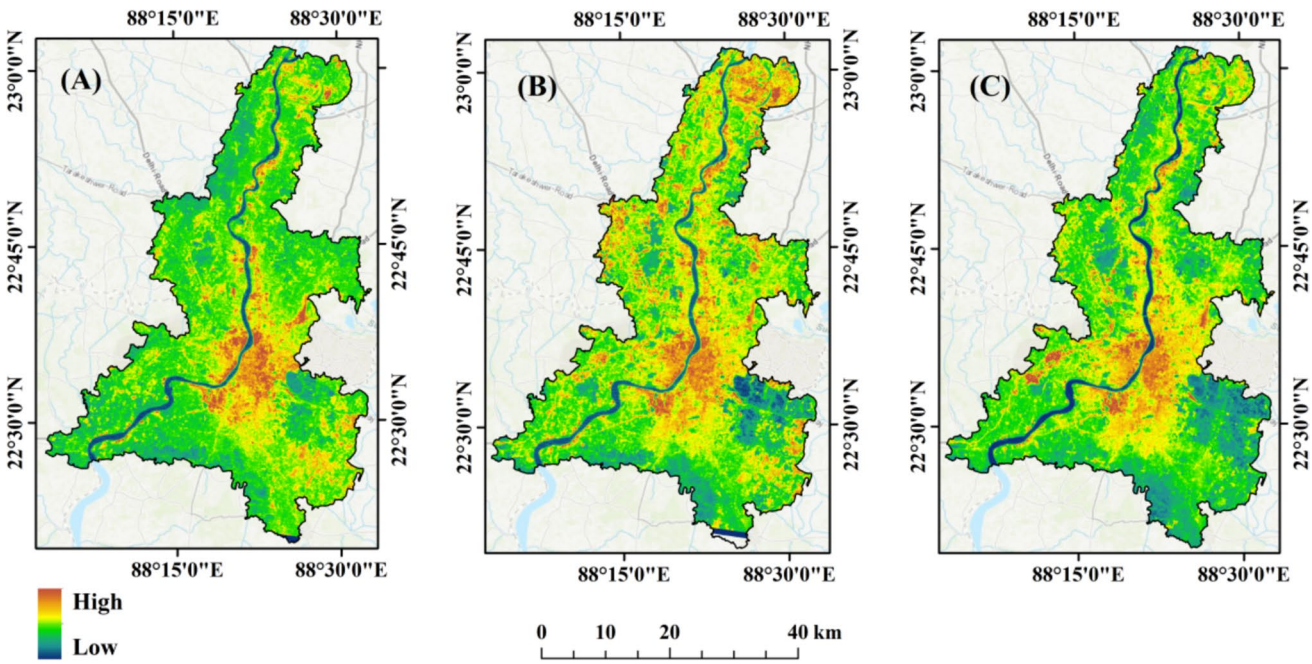

Fig. 5 Spatio-temporal pattern of LST maps (**A**) 2000, (**B**) 2010, and (**C**) 2020. LST means land surface temperature. LST refers to the land surface temperature

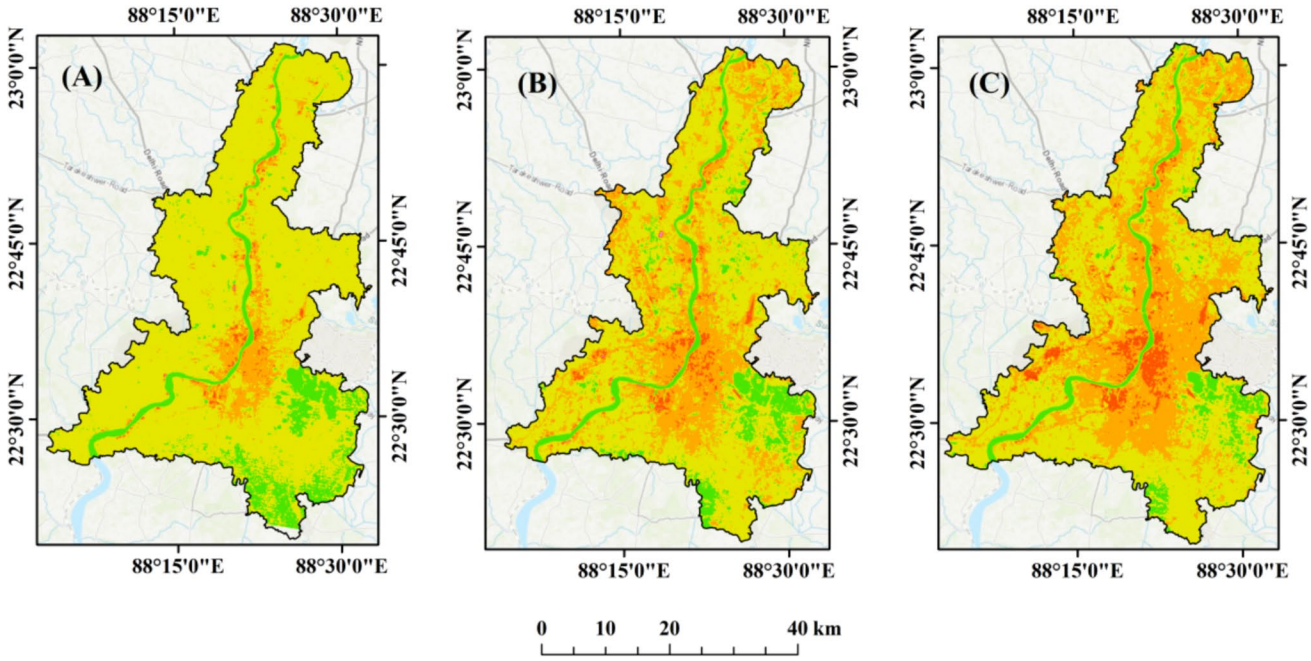

Fig. 6 Spatio-temporal pattern of SUHI (**A**) 2000, (**B**) 2010, and (**C**) 2020. SUHI means surface urban heat island. SUHI refers to the surface urban heat island

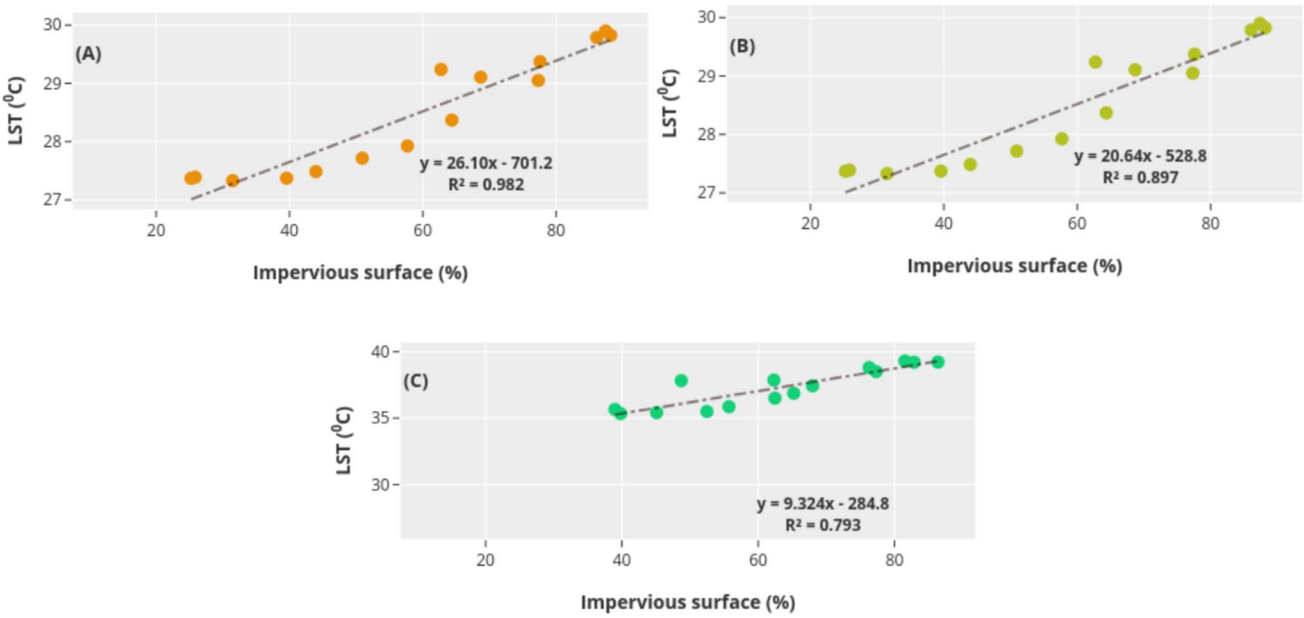

Fig. 7 Relationship between LST and IS (%) ((**A**) 2000, (**B**) 2010, and (**C**) 2020. LST and IS means land surface temperature and impervious surface. LST refers to the land surface temperature and IS refers to the impervious surface

UHI intensity significantly increased due to the rapid transformation of the previous surface into an IS area (Mathew et al. 2017). For example, Siddiqui et al. (2021) carried out a study of large cities in India to understand the UHI intensity, and their findings showed that spatial extension of UHI intensity increased. One of the prime drivers of UHI increase is the conversion of the previous surface into the IS, for example, the conversion of water bodies, vegetation cover, and wetland areas into built-up areas (Mohammad et al. 2022; Ramzan et al. 2022). In this study, it was observed

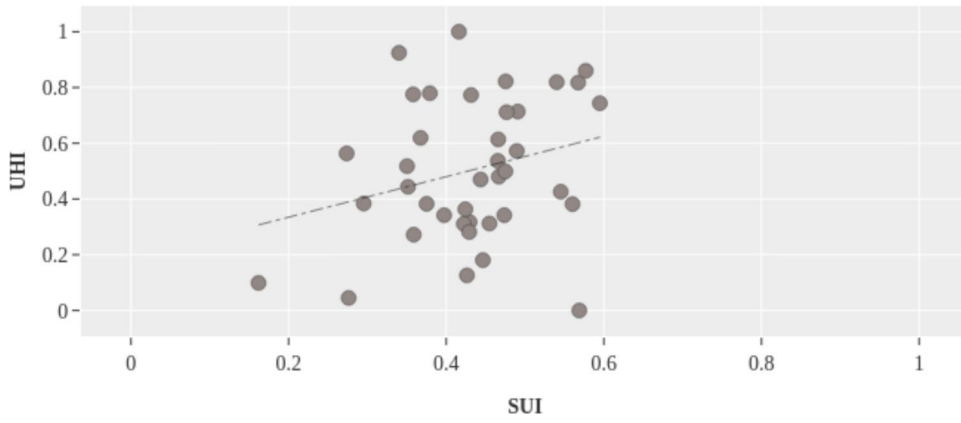

Fig. 8 Relationship between SUHI and SUI (2020). SUHI and SUI means surface urban heat island and spatial urbanization index. SUHI refers to the surface urban heat island and SUI refers to the spatial urbanization index

Table 2 Quantitative metrics of spatial urbanization factors (RF regression). RF stands for random forest

Statistics	Results
MSE	0.28
RMSE	0.53
MAE/MAD	0.40
MAPE	53.0%
R^2	0.87

NB: MSE-mean square error, RMSE-root mean square error, MAE-mean absolute error, MAPE-mean absolute percentage error.

that the areas with higher vegetation cover, water bodies, and built-up areas showed higher UHI intensity denoting the fact that socio-ecological variables have a significant impact on UHI intensity (Zhang et al. 2017a, b).

The regression analysis shows a clear relationship between IS and LST over the 20-year period from 2000 to 2020. In 2000, the data indicate a strong correlation ($R^2 = 0.982$), where a 1% increase in IS leads to a significant 26.10 °C rise in LST. This reflects how early urbanization contributed heavily to UHI effect due to extensive IS coverage, which limits natural cooling processes like

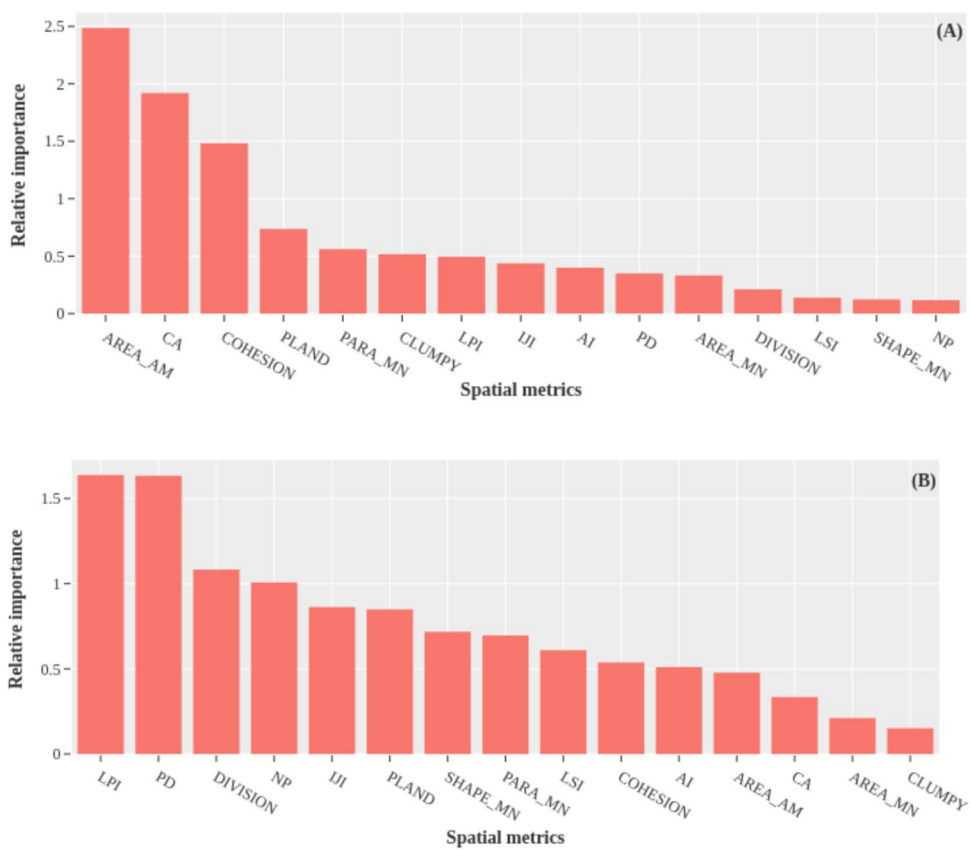

Fig. 9 (**A**) Relative importance of landscape metrics on spatial urbanization (**B**) relative importance of spatial urbanization metrics on SUHI. SUHI refers to the surface urban heat island. AREA_AM- area-weighted mean patch area; CA- class area; PLAND- percentage of landscape; PARA_MN- perimeter area mean; LPI- largest patch index; IJI- interspersion and juxtaposition index; AI- aggregation index; PD- patch density; AREA_MN- area patch mean; LSI- landscape shape index; SHAPE_MN- shape mean; NP- number of patches

Table 3 Quantitative metrics of spatial metrics on thermal environment (RF regression). RF means random forest

Statistics	Results
MSE	0.511
RMSE	0.715
MAE	0.571
MAPE	143%
R^2	0.483

NB: MSE-mean square error, RMSE-root mean square error, MAE-mean absolute error, MAPE-mean absolute percentage error.

evapotranspiration (Wang et al. 2019; Li et al. 2020). By 2010, although the relationship remains robust ($R^2 = 0.897$), the LST increase per 1% IS drops to 20.64 °C. This may be linked to early efforts in urban planning that incorporate green infrastructure and climate-responsive designs, which partially mitigate the UHI effect (Zhao et al. 2018). By 2020, the correlation weakens further ($R^2 = 0.793$), with a 9.324 °C rise per 1% IS. This suggests that urban areas may have adopted advanced mitigation strategies such as increased vegetation cover, cool roofs, and reflective materials, which reduce surface heating (Gunawardena et al. 2017). Additionally, climate change adaptation policies may have contributed to moderating the relationship between IS and LST (Yang et al. 2021). The gradual decline in the LST response to IS suggests a combination of evolving urban infrastructure and improved environmental practices (Wu et al. 2022).

The relative importance of spatial metrics in influencing spatial urbanization and UHI dynamics is critical for understanding urban development and environmental impacts. In this study, RF regression models were used to assess the significance of various spatial metrics on both spatial urbanization and UHI. The model for spatial urbanization yielded an R^2 value of 0.87, indicating that 87% of the variance in urbanization patterns was explained by fifteen spatial metrics. AREA_AMemerged as the most influential metric, suggesting that larger patches of built-up areas strongly drive urbanization. Other metrics such as CA, COHESION, PLAND, PARA_MN, CLUMPY, and LPI also played significant roles in shaping urban patterns (Buyantuyev and Wu 2010; Herold et al. 2003). In the context of UHI, the RF model explained 48% of the variance in the thermal environment, with an R^2 value of 0.48. This lower explanatory power suggests that while spatial metrics contribute to UHI, other factors, such as surface materials and vegetation, also play crucial roles (Imhoff et al. 2010). Among the fourteen spatial metrics evaluated, LPI had the greatest influence on UHI, highlighting the impact of large, contiguous urban patches on heat accumulation (Oke 1982; Peng et al. 2016). Other significant metrics included PD, DIVISION, NP (number of patches), and IJI (interspersion and juxtaposition index), all of which indicate that fragmentation and spatial configuration of urban landscapes affect heat distribution (Jenerette et al. 2007; Li et al. 2011).These results underscore the critical role of spatial metrics in explaining urbanization and UHI dynamics, emphasizing the need for urban planners to consider spatial patterns when addressing UHI mitigation strategies (Li et al. 2021). Metrics such as LPI and CLUMPY could be targeted for developing effective interventions to reduce SUHI and manage sustainable urban expansion (Wu et al. 2014; Yu et al. 2018).

Policy implications

In developing countries, the increase of surface as well as atmospheric temperature has become one of the serious environmental challenges. Therefore, it is essential to implement effective landscape design and urban planning. The observed surface, as well as atmospheric temperature, significantly increased than predicted across the globe, and it is expected to increase further in next decades (Leng et al. 2015; Pal and Eltahir 2016). The variation of temperature within the cities provides significant hints for better future urban design and landscape design. The cities play a significant role in resolving the UHI effect within the cities through nature-based solutions (NBS) (such as green spaces restoration and enhancement). Thus, it provides opportunities to mitigate the UHI effect in cities. Thus, nature-based strategies provide a great opportunity to mitigate the UHI effect. A number of mitigation strategies were proposed in previous studies to solve the UHI effect in urban areas, such as an increase in vegetation cover, creation of green roofs, application of cool pavements and conservation of natural landscapes in urban areas (Grover and Singh 2016). Although a number of measures were adopted to mitigate the UHI effect, green infrastructure (GI) can be implemented easily in any urban area to mitigate the UHI effect (Marandoet al. 2019; Balanyet al. 2020). Green space in an urban area helps in reducing surface and atmospheric temperature (Zhanget al. 2017a, b; Alavipanahet al. 2015). Thus, GI can be utilized in urban areas by enhancing vegetation cover, green walls, and street trees (Heet al. 2021; Imranet al. 2019). Thus, a NBS has been recognized as the key solution to mitigating the UHI effect in developing countries and large and small cities (Ferrario et al. 2024). Though NBS has emerged as one of the strategic solutions to mitigate UHI in cities of developing, still it is very challenging as cities are highly compacted and dense like Kolkata. Thus, strategic urban planning is necessary to implement NBS in India's compact and densely constructed cities in order to mitigate the UHI effect. Initially, the coverage of vegetation in restricted areas can be improved by vertical greening, which involves the use of green walls, rooftop gardens, and terrace farming.

Secondly, the integration of street trees along narrow streets should be prioritized, with a focus on native, drought-resistant species that provide shelter and cooling. Third, vacant sites, railway corridors, and degraded lands can be transformed into urban forests and pocket parks to improve green infrastructure. Fourth, surface heating should be mitigated and evapotranspiration should be improved by incorporating blue-green infrastructure, including urban wetlands, rain gardens, and bioswales. Fifth, reflective pavements and cold roofing can mitigate heat absorption. Lastly, policy interventions, including the implementation of green building standards, the promotion of green infrastructure, and community-driven afforestation, are indispensable. An approach that is multi-scale, participatory, and involves policymakers, urban planners, and citizens is essential for the development of sustainable and climate-resilient Indian cities. Through, NBS is crucial to mitigate UHI, but, the cities of the developing countries experience many challenges. The implementation of NBS to mitigate UHI effect in cities of the Global South may face numerous obstacles. For example, large-scale green infrastructure may constrain by space constraints in densely populated locations. Municipal budgets may burden by the high maintenance costs of permeable surfaces, urban forests, and green roofs. The survival of trees and the efforts to restore urban areas may influence by water scarcity. Poor coordination between government agencies may be a consequence of institutional fragmentation. Adoption may impede by a lack of public awareness and community participation. The implementation of large-scale projects may impede by the limited infrastructure of informal settlements. NBS initiatives are frequently overshadowed by competing urban priorities, such as housing and transport, necessitating integrated, multi-sectoral planning to be effective.

Thus, the application of a NBS is not only important to mitigate UHI rather achieve sustainable development goals (SDGs) (Fig. 10). The rapid transformation of agricultural land into other landscapes in the cities of developing countries significantly affects food production. Thus, sustainable urban planning is urgently required to achieve Goal 2 (No Hunger- SDG 2). The increase in UHI effect negatively affects human health and triggers heat stress and thermal discomfort (Luo and Lau 2021). Therefore, it is essential to achieve 'Good Health and Well-being' (SDG 3). GS helps to reduce energy consumption and save the energy specially at the domestic and commercial scale. Thus, it would ensure access to affordable and sustainable energy to achieve SDG 7 ('Affordable and Clean Energy'). GS plays significant role for cooling, to reduce pollutions and enhance well-being of the people (Zhang et al. 2014; Nath et al. 2018). Thus, it ensures and build a resilient city and liable for the communities (SDG 11- 'Sustainable Cities and Communities'). Climate change has emerged as one of the serious challenges across the globe (Capstick et al. 2015). Therefore, it is essential to utilize a natural-based solution to mitigate the UHI effect through climate-related actions (Climate Actions-SDG 13). Finally, the utilization of effective management strategies to mitigate the UHI effect is essential to sustain the well-being and livelihoods of the urban dwellers and to achieve 'Life on the Land' (SDG 15).

Limitations and future research directions

The study provides significant findings to understand the relationship between urbanization and UHI. Still, there are a few limitations of the study. Firstly, UHI has been assessed based on the year 2000, 2010 and 2020. No time series analysis was carried out. Therefore, there is a scope to study series analysis in the future. Secondly, UHI varies seasonally i.e. there is a variation of the UHI in summer and winter. Therefore, future researchers should consider the seasonal analysis of UHI in KMA for a better understanding of the spatial variation. Lastly, there are many approaches to measuring urbanization such as population, social, economic and spatial. In this study, only spatial urbanization has been measured based on the availability of data related to the landscape metrics. This study suggests measuring urbanization from all perspectives as much as possible for better analysis and understanding the spatial scenario of urbanization. Despite, the limitations, the findings of the study are crucial to understanding the relationship between urbanization and UHI.

Conclusions

This study effectively highlights the significant impact of spatial urbanization on UHI intensity using a machine learning-based landscape approach. The findings reveal a substantial increase in IS and a decline in natural ecosystems, such as vegetation cover, water bodies, and agricultural lands, from 2000 to 2020. These land cover changes have directly contributed to an increase in urban heat stress, with LPI emerging as the most influential factor affecting UHI intensity according to the RF model. The results underscore the urgent need for sustainable urban planning strategies to mitigate the adverse effects of urban expansion on thermal environments.

To effectively combat UHI, integrating green and blue infrastructure into urban planning policies is crucial. However, several challenges must be addressed, including governance constraints, funding limitations, and public acceptance of NBS. Strong institutional frameworks and multi-stakeholder collaboration are required to ensure effective implementation. Moreover, financial investments in urban greening initiatives should be prioritized, supported by incentives such as tax benefits for green roofs,

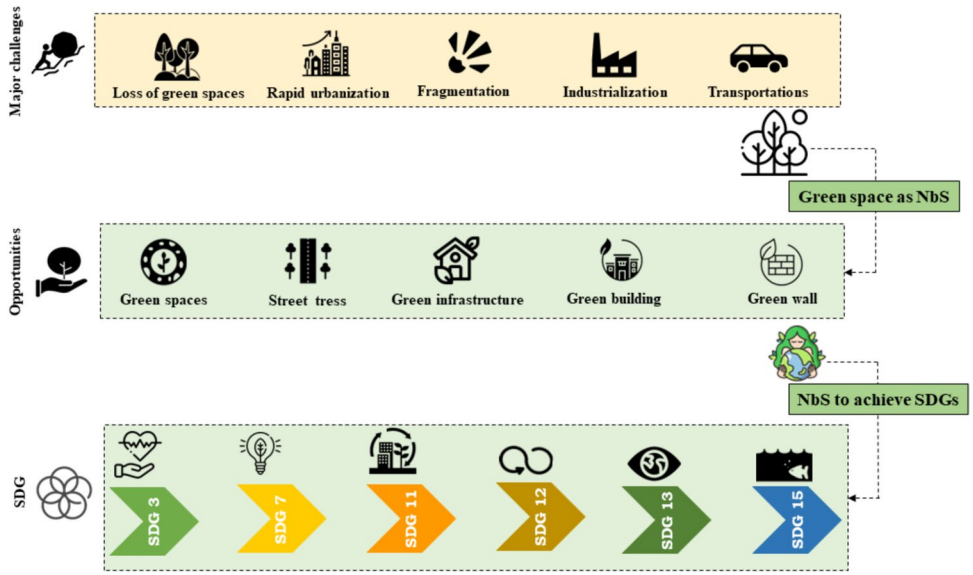

Fig. 10 Potential paths to mitigate heat in cities and achieve SDGs. SDG refers to the Sustainable Development Goals

subsidies for urban forestry, and public–private partnerships to promote sustainable infrastructure development. Additionally, raising awareness and engaging communities in the decision-making process can enhance public acceptance and long-term success of UHI mitigation strategies. Beyond immediate temperature regulation, urban greening plays a critical role in strengthening urban resilience in the context of climate change adaptation. The incorporation of green infrastructure not only mitigates UHI but also enhances biodiversity, improves air quality, and reduces flood risks, contributing to multiple SDGs. Specifically, these measures align with SDG 11 (Sustainable Cities and Communities) by improving urban livability and SDG 13 (Climate Action) by promoting climate resilience. Urban green spaces also support SDG 3 (Good Health and Well-being) by providing cooling effects, reducing heat-related illnesses, and enhancing mental well-being.

Thus, this study emphasizes the necessity of sustainable urban design to counteract the negative effects of urbanization on thermal environments. Implementing comprehensive greening strategies, supported by policy reforms, financial mechanisms, and community engagement, can foster climate-resilient cities. A proactive approach integrating NBS within urban planning frameworks will be essential in creating more sustainable and livable cities in the face of ongoing climate challenges.

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Data availability The datasets are available on reasonable request.

Declarations

Ethical approval Not applicable

Consent to participate Not Applicable

Consent to publish Not Applicable

Clinical trial number Not Applicable

Human ethics and consent to participate Not Applicable

Competing interests The authors declare that they have no competing interests.

References

Adeyemi A, Ramoelo A, Cho MA, Strydom J (2021) Spatio-temporal analysis of built-up impervious surface area and interplay with land surface temperature in Pretoria, South Africa. Geocarto Intl 1–21

Alavipanah SK, Hashemi Darrehbadami S, Kazemzadeh A (2015) Spatial-temporal analysis of urban heat-island of Mashhad city due to land use/cover change and expansion. Geograph Urban Plann Res (GUPR) 3(1):1–17

Almeida CRD, Teodoro AC, Gonçalves A (2021) Study of the urban heat island (UHI) using remote sensing data/techniques: a systematic review. Environments 8(10):105

Arshad A, Ashraf M, Sundari RS, Qamar H, Wajid M, Hasan MU (2020) Vulnerability assessment of urban expansion and modelling green spaces to build heat waves risk resiliency in Karachi. Int J Disaster Risk Reduct 46:101468. https://doi.org/10.1016/j.ijdrr.2019.101468

Balany F, Ng AW, Muttil N, Muthukumaran S, Wong MS (2020) Green infrastructure as an urban heat island mitigation strategy—a review. Water 12(12):3577

Bai Y, Guo R, Xing Y (2018) Relationship between urban heat island and green infrastructure fraction in Harbin. In: Remote sensing technologies and applications in urban environments III, vol 10793, pp 54–66). SPIE. https://doi.org/10.1117/12.2501937

Bharath HA, Chandan MC, Vinay S, Ramachandra TV (2018) Modelling urban dynamics in rapidly urbanising Indian cities. Egypt J Remote Sens Space Sci 21(3):201–210

Buyantuyev A, Wu J (2010) Urban heat islands and landscape heterogeneity: linking spatiotemporal variations in surface temperatures to land-cover and socioeconomic patterns. Landsc Ecol 25:17–33

Capstick S, Whitmarsh L, Poortinga W, Pidgeon N, Upham P (2015) International trends in public perceptions of climate change over the past quarter century. WIREs Clim Change 6(1):35–61

Chatterjee S, Dinda A (2022) Determination of characterized urban thermal zones (UTZ) for assessing microclimates in the tropical metropolitan area of Kolkata. Sustain Cities Soc 80:103807. https://doi.org/10.1016/j.scs.2022.103807

Chatterjee U, Majumdar S (2022) Impact of land use change and rapid urbanization on urban heat island in Kolkata city: a remote sensing based perspective. J Urban Manag 11(1):59–71

Chen G, Chen Y, He H, Wang J, Zhao L, Cai Y (2023) Assessing the synergies between heat waves and urban heat islands of different local climate zones in Guangzhou, China. Build Environ 240:110434. https://doi.org/10.1016/j.buildenv.2023.110434

Chen X, Xu Y, Yang J, Wu Z, Zhu H (2020) Remote sensing of urban thermal environments within local climate zones: a case study of two high-density subtropical Chinese cities. Urban Clim 31:100568. https://doi.org/10.1016/j.uclim.2019.100568

Chen Y, Shan B, Yu X (2022) Study on the spatial heterogeneity of urban heat islands and influencing factors. Build Environ 208:108604. https://doi.org/10.1016/j.buildenv.2021.108604

Deilami K, Kamruzzaman M, Liu Y (2018) Urban heat island effect: a systematic review of spatio-temporal factors, data, methods, and mitigation measures. Int J Appl Earth Obs Geoinf 67:30–42

Deng C, Wu C (2012) BCI: a biophysical composition index for remote sensing of urban environments. Remote Sens Environ 127:247–259

Dewan A, Kiselev G, Botje D, Mahmud GI, Bhuian MH, Hassan QK (2021) Surface urban heat island intensity in five major cities of Bangladesh: patterns, drivers and trends. Sustain Cities Soc 71:102926. https://doi.org/10.1016/j.scs.2021.102926

Dutta K, Basu D, Agrawal S (2022) Evaluation of seasonal variability in magnitude of urban heat islands using local climate zone classification and surface albedo. Int J Environ Sci Technol 19(9):8677–8698

Estoque RC, Murayama Y, Myint SW (2017) Effects of landscape composition and pattern on land surface temperature: an urban heat island study in the megacities of Southeast Asia. Sci Total Environ 577:349–359

Ferrario F, Mourato JM, Rodrigues MS, Dias LF (2024) Evaluating nature-based solutions as urban resilience and climate adaptation tools: a meta-analysis of their benefits on heatwaves and floods. Sci Total Environ. https://doi.org/10.1016/j.scitotenv.2024.175179

Ferrini F, Fini A, Mori J, Gori A (2020) Role of vegetation as a mitigating factor in the urban context. Sustainability 12(10):4247

Ghenai C, Rejeb O, Sinclair T, Almarzouqi N, Alhanaee N, Rossi F (2023) Evaluation and thermal performance of cool pavement under desert weather conditions: surface albedo enhancement and carbon emissions offset. Case Stud Constr Mater 18:e01940

Grover A, Singh RB (2016) Monitoring spatial patterns of land surface temperature and urban heat island for sustainable megacity: a case study of Mumbai, India, using Landsat TM data. Environ Urbanization Asia 7(1):38–54

Gunawardena KR, Wells MJ, Kershaw T (2017) Utilising green and bluespace to mitigate urban heat island intensity. Sci Total Environ 584:1040–1055

Gupta N, Mathew A, Khandelwal S (2019) Analysis of cooling effect of water bodies on land surface temperature in nearby region: a case study of Ahmedabad and Chandigarh cities in India. Egypt J Remote Sens Space Sci 22(1):81–93

Halder B, Bandyopadhyay J, Banik P (2021) Monitoring the effect of urban development on urban heat island based on remote sensing and geo-spatial approach in Kolkata and adjacent areas, India. Sustain Cities Soc 74:103186. https://doi.org/10.1016/j.scs.2021.103186

He BJ, Wang J, Liu H, Ulpiani G (2021) Localized synergies between heat waves and urban heat islands: implications on human thermal comfort and urban heat management. Environ Res 193:110584. https://doi.org/10.1016/j.envres.2020.110584

Herold M, Liu X, Clarke KC (2003) Spatial metrics and image texture for mapping urban land use. Photogramm Eng Remote Sens 69(9):991–1001

Hua L, Zhang X, Nie Q, Sun F, Tang L (2020) The impacts of the expansion of urban impervious surfaces on urban heat islands in a coastal city in China. Sustainability 12(2):475

Huang WTK, Masselot P, Bou-Zeid E, Fatichi S, Paschalis A, Sun T, Manoli G (2023) Economic valuation of temperature-related mortality attributed to urban heat islands in European cities. Nat Commun 14(1):7438

Imhoff ML, Zhang P, Wolfe RE, Bounoua L (2010) Remote sensing of the urban heat island effect across biomes in the continental USA. Remote Sens Environ 114(3):504–513

Imran HM, Kala J, Ng AWM, Muthukumaran S (2019) Effectiveness of vegetated patches as green infrastructure in mitigating urban heat island effects during a heatwave event in the city of Melbourne. Weather Clim Extremes 25:100217. https://doi.org/10.1016/j.wace.2019.100217

Irfeey AMM, Chau HW, Sumaiya MMF, Wai CY, Muttil N, Jamei E (2023) Sustainable mitigation strategies for urban heat island effects in urban areas. Sustainability 15(14):10767

Jain M, Dawa D, Mehta R, Dimri AP, Pandit MK (2016) Monitoring land use change and its drivers in Delhi, India using multi-temporal satellite data. Model Earth Syst Environ 2(1):1–14

Jenerette GD, Harlan SL, Brazel A, Jones N, Larsen L, Stefanov WL (2007) Regional relationships between surface temperature, vegetation, and human settlement in a rapidly urbanizing ecosystem. Landsc Ecol 22:353–365

Jin H, Cui P, Wong NH, Ignatius M (2018) Assessing the effects of urban morphology parameters on microclimate in Singapore to control the urban heat island effect. Sustainability 10(1):206

Kong F, Yin H, James P, Hutyra LR, He HS (2014) Effects of spatial pattern of greenspace on urban cooling in a large metropolitan area of eastern China. Landsc Urban Plann 128:35–47

Kong J, Zhao Y, Carmeliet J, Lei C (2021) Urban heat island and its interaction with heatwaves: a review of studies on mesoscale. Sustainability 13(19):10923

Kotharkar R, Bagade A (2018) Evaluating urban heat island in the critical local climate zones of an Indian city. Landsc Urban Plann 169:92–104

Laue F, Adegun OB, Ley A (2022) Heat stress adaptation within informal, low-income urban settlements in Africa. Sustainability 14(13):8182

Leng G, Tang Q, Rayburg S (2015) Climate change impacts on meteorological, agricultural and hydrological droughts in China. Glob Planet Change 126:23–34

Li F, Yigitcanlar T, Li W, Nepal M, Nguyen K, Dur F (2024) Understanding urban heat vulnerability: scientometric analysis of five decades of research. Urban Clim 56:102035. https://doi.org/10.1016/j.uclim.2024.102035

Li H, Meier F, Lee X, Chakraborty T, Liu J, Schaap M, Sodoudi S (2018) Interaction between urban heat island and urban pollution island during summer in Berlin. Sci Total Environ 636:818–828

Li H, Li L, Chen L, Zhou X, Cui Y, Liu Y, Liu W (2019) Mapping and characterizing spatiotemporal dynamics of impervious surfaces using Landsat images: a case study of Xuzhou, East China from 1995 to 2018. Sustainability 11(5):1224

Li J, Song C, Cao L, Zhu F, Meng X, Wu J (2011) Impacts of landscape structure on surface urban heat islands: a case study of Shanghai, China. Remote Sens Environ 115(12):3249–3263

Li X, Stringer LC, Dallimer M (2021) The spatial and temporal characteristics of urban heat island intensity: implications for East Africa's urban development. Climate 9(4):51

Li Y, Sun Y, Li J, Gao C (2020) Socioeconomic drivers of urban heat island effect: empirical evidence from major Chinese cities. Sustain Cities Soc 63:102425. https://doi.org/10.1016/j.scs.2020.102425

Li Z, Xie M, Wang H, Chen B, Wu R, Chen Y (2022) The spatiotemporal heterogeneity of the relationship between PM2. 5 concentrations and the surface urban heat island effect in Beijing, China. Prog Phys Geogr Earth Environ 46(1):84–104

Lin Y, Wang Z, Jim CY, Li J, Deng J, Liu J (2020) Water as an urban heat sink: blue infrastructure alleviates urban heat island effect in mega-city agglomeration. J Clean Prod 262:121411. https://doi.org/10.1016/j.jclepro.2020.121411

Lin J, Zhang H, Chen M, Wang Q (2023) Socioeconomic disparities in cooling and warming efficiencies of urban vegetation and impervious surfaces. Sustain Cities Soc 92:104464. https://doi.org/10.1016/j.scs.2023.104464

Lu Y, Qin Y, Huang C, Pang X (2023) Albedo of pervious concrete and its implications for mitigating urban heat island. Sustainability 15(10):8222

Luo M, Lau NC (2021) Increasing human-perceived heat stress risks exacerbated by urbanization in China: a comparative study based on multiple metrics. Earths Future 9(7):e2020EF001848

Mandal J, Patel PP, Samanta S (2022) Examining the expansion of urban heat island effect in the Kolkata Metropolitan Area and its vicinity using multi-temporal MODIS satellite data. Adv Space Res 69(5):1960–1977

Marando F, Heris MP, Zulian G, Udías A, Mentaschi L, Chrysoulakis N, Maes J (2022) Urban heat island mitigation by green infrastructure in European Functional Urban Areas. Sustain Cities Soc 77:103564. https://doi.org/10.1016/j.scs.2021.103564

Marando F, Salvatori E, Sebastiani A, Fusaro L, Manes F (2019) Regulating ecosystem services and green infrastructure: assessment of urban heat island effect mitigation in the municipality of Rome, Italy. Ecol Modell 392:92–102

Mathew A, Khandelwal S, Kaul N (2017) Investigating spatial and seasonal variations of urban heat island effect over Jaipur city and its relationship with vegetation, urbanization and elevation parameters. Sustain Cities Soc 35:157–177

McGarigal K, Marks BJ (1995) FRAGSTATS: spatial pattern analysis program for quantifying landscape structure. In: USDA forest service general technical report PNW-351, Corvallis

McGarigal K, Cushman SA (2002) Comparative evaluation of experimental approaches to the study of habitat fragmentation effects. Ecol Appl 12(2):335–345

Meng Q, Zhang L, Sun Z, Meng F, Wang L, Sun Y (2018) Characterizing spatial and temporal trends of surface urban heat island effect in an urban main built-up area: a 12-year case study in Beijing, China. Remote Sens Environ 204:826–837

Miah MT, Fariha JN, Kafy AA, Islam R, Biswas N, Duti BM, Salem MA (2024) Exploring the nexus between land cover change dynamics and spatial heterogeneity of demographic trajectories in rapidly growing ecosystems of south Asian cities. Ecol Indic 158:111299. https://doi.org/10.1016/j.ecolind.2023.111299

Mirabi E, Davies PJ (2022) A systematic review investigating linear infrastructure effects on urban heat island (UHIULI) and its interaction with UHI typologies. Urban Clim 45:101261. https://doi.org/10.1016/j.uclim.2022.101261

Mohajerani A, Bakaric J, Jeffrey-Bailey T (2017) The urban heat island effect, its causes, and mitigation, with reference to the thermal properties of asphalt concrete. J Environ Manage 197:522–538

Mohammad P, Goswami A, Chauhan S, Nayak S (2022) Machine learning algorithm based prediction of land use land cover and land surface temperature changes to characterize the surface urban heat island phenomena over Ahmedabad city, India. Urban Climate 42:101116. https://doi.org/10.1016/j.uclim.2022.101116

Morabito M, Crisci A, Guerri G, Messeri A, Congedo L, Munafò M (2021) Surface urban heat islands in Italian metropolitan cities: tree cover and impervious surface influences. Sci Total Environ 751:142334. https://doi.org/10.1016/j.scitotenv.2020.142334

Mukherjee M (2012) Urban growth and spatial transformation of Kolkata metropolis: a continuation of colonial legacy. ARPN J Sci Technol 2(3):365–380

Mukhopadhyay B, Weitz CA (2022) Heat exposure, heat-related symptoms and coping strategies among elderly residents of urban slums and rural vilages in West Bengal, India. Int J Environ Res Public Health 19(19):12446

Nath TK, Han SSZ, Lechner AM (2018) Urban green space and well-being in Kuala Lumpur, Malaysia. Urban Forestry Urban Greening 36:34–41

Oke TR (1982) The energetic basis of the urban heat island. Q J R Meteorol Soc 108(455):1–24

Oke TR, Cleugh HA (1987) Urban heat storage derived as energy balance residuals. Bound-Layer Meteorol 39(3):233–245

Pal JS, Eltahir EA (2016) Future temperature in southwest Asia projected to exceed a threshold for human adaptability. Nat Clim Chang 6(2):197–200

Patel P, Karmakar S, Ghosh S, Aliaga DG, Niyogi D (2021) Impact of green roofs on heavy rainfall in tropical, coastal urban area. Environ Res Lett 16(7):074051. https://doi.org/10.1088/1748-9326/ac1011

Peng J, Jia J, Liu Y, Li H, Wu J (2018) Seasonal contrast of the dominant factors for spatial distribution of land surface temperature in urban areas. Remote Sens Environ 215:255–267

Peng J, Tian L, Liu Y, Zhao M, Hu YN, Wu J (2017) Ecosystem services response to urbanization in metropolitan areas: thresholds identification. Sci Total Environ 607:706–714

Peng J, Xie P, Liu Y, Ma J (2016) Urban thermal environment dynamics and associated landscape pattern factors: a case study in the Beijing metropolitan region. Remote Sens Environ 173:145–155

Qiao K, Zhu W, Hu D, Hao M, Chen S, Cao S (2018) Examining the distribution and dynamics of impervious surface in different function zones in Beijing. J Geogr Sci 28(5):669–684

Rahman MN, Rony MRH, Jannat FA, Chandra Pal S, Islam MS, Alam E, Islam ARMT (2022) Impact of urbanization on urban heat island intensity in major districts of Bangladesh using remote sensing and geo-spatial tools. Climate. https://doi.org/10.3390/cli10010003

Ramakreshnan L, Aghamohammadi N, Fong CS, Ghaffarianhoseini A, Wong LP, Noor RM, Hassan N (2019) A qualitative exploration on the awareness and knowledge of stakeholders towards urban heat island phenomenon in Greater Kuala Lumpur: critical insights for urban policy implications. Habitat Int 86:28–37

Ramzan M, Saqib ZA, Hussain E, Khan JA, Nazir A, Dasti MYS, Niazi NK (2022) Remote sensing-based prediction of temporal changes in land surface temperature and land use-land cover (LULC) in urban environments. Land 11(9):1610

Rashid SF, Gani S, Sarker M (2013) Urban poverty, climate change and health risks for slum dwellers in Bangladesh. Climate change adaptation actions in Bangladesh 51–70

Rihan M, Naikoo MW, Ali MA, Usmani TM, Rahman A (2021) Urban heat island dynamics in response to land-use/land-cover change in the coastal city of Mumbai. J Indian Soc Remote Sens 49(9):2227–2247

Roy MB, Ghosh A, Mohinuddin S, Kumar A, Roy PK (2022) Analysing the trending nature in land surface temperature on different land use land cover changes in urban lakes, West Bengal, India. Model Earth Syst Environ 8(4):4603–4627

Siddiqui A, Kushwaha G, Nikam B, Srivastav SK, Shelar A, Kumar P (2021) Analysing the day/night seasonal and annual changes and trends in land surface temperature and surface urban heat island intensity (SUHII) for Indian cities. Sustain Cities Soc 75:103374. https://doi.org/10.1016/j.scs.2021.103374

Sun D, Hu C, Wang Y, Wang Z, Zhang J (2021) Examining spatio-temporal characteristics of urban heat islands and factors driving them in Hangzhou, China. IEEE J Sel Top Appl Earth Obs Remote Sens 14:8316–8325

Wang J, Zhou W (2022) More urban greenspace, lower temperature? Moving beyond net change in greenspace. Agric for Meteorol 322:109021. https://doi.org/10.1016/j.agrformet.2022.109021

Wang J, Zhou W, Wang J (2019) Time-series analysis reveals intensified urban heat island effects but without significant urban warming. Remote Sens 11(19):2229

Wang Y, Li X, Zhang C, He W (2022) Influence of spatiotemporal changes of impervious surface on the urban thermal environment: a case of Huai'an central urban area. Sustain Cities Soc 79:103710. https://doi.org/10.1016/j.scs.2022.103710

Wu J (2013) Landscape sustainability science: ecosystem services and human well-being in changing landscapes. Landsc Ecol 28(6):999–1023

Wu H, Ye LP, Shi WZ, Clarke KC (2014) Assessing the effects of land use spatial structure on urban heat islands using HJ-1B remote sensing imagery in Wuhan, China. Int J Appl Earth Obs Geoinf 32:67–78

Wu X, Wang G, Yao R, Wang L, Yu D, Gui X (2019) Investigating surface urban heat islands in South America based on MODIS data from 2003–2016. Remote Sens 11(10):1212

Wu Y, Hou H, Wang R, Murayama Y, Wang L, Hu T (2022) Effects of landscape patterns on the morphological evolution of surface urban heat island in Hangzhou during 2000–2020. Sustain Cities Soc 79:103717. https://doi.org/10.1016/j.scs.2022.103717

Xu H, Lin D, Tang F (2013) The impact of impervious surface development on land surface temperature in a subtropical city: Xiamen China. Int J Climatol 33(8):1873–1883

Xu J, Zhao Y, Zhong K, Zhang F, Liu X, Sun C (2018) Measuring spatio-temporal dynamics of impervious surface in Guangzhou, China, from 1988 to 2015, using time-series Landsat imagery. Sci Total Environ 627:264–281

Yang G, Ren G, Zhang P, Xue X, Tysa SK, Jia W, Zhang S (2021) PM2. 5 influence on urban heat island (UHI) effect in Beijing and the possible mechanisms. J Geophys Res Atmos 126(17):e2021JD035227

Yao L, Li T, Xu M, Xu Y (2020a) How the landscape features of urban green space impact seasonal land surface temperatures at a city-block-scale: an urban heat island study in Beijing, China. Urban For Urban Green 52:126704. https://doi.org/10.1016/j.ufug.2020.126704

Yao N, Huang C, Yang J, Konijnendijk van den Bosch CC, Ma L, Jia Z (2020b) Combined effects of impervious surface change and large-scale afforestation on the surface urban heat island intensity of Beijing, China based on remote sensing analysis. Remote Sens 12(23):3906

Yu Z, Guo X, Zeng Y, Koga M, Vejre H (2018) Variations in land surface temperature and cooling efficiency of green space in rapid urbanization: the case of Fuzhou City, China. Urban For Urban Green 29:113–121

Zhang B, Gao JX, Yang Y (2014) The cooling effect of urban green spaces as a contribution to energy-saving and emission-reduction: a case study in Beijing, China. Build Environ 76:37–43

Zhang H, Qi ZF, Ye XY, Cai YB, Ma WC, Chen MN (2013) Analysis of land use/land cover change, population shift, and their effects on spatiotemporal patterns of urban heat islands in metropolitan Shanghai, China. Appl Geogr 44:121–133

Zhang W, Li Y, Zheng C, Zhu Y (2023) Surface urban heat island effect and its driving factors for all the cities in China: based on a new batch processing method. Ecol Indic 146:109818. https://doi.org/10.1016/j.ecolind.2022.109818

Zhang X, Estoque RC, Murayama Y (2017a) An urban heat island study in Nanchang City, China based on land surface temperature and social-ecological variables. Sustain Cities Soc 32:557–568

Zhang Y, Sun L (2019) Spatial-temporal impacts of urban land use land cover on land surface temperature: case studies of two Canadian urban areas. Int J Appl Earth Obs Geoinf 75:171–181

Zhang Y, Li D, Liu L, Liang Z, Shen J, Wei F, Li S (2021) Spatiotemporal characteristics of the surface urban heat island and its driving factors based on local climate zones and population in Beijing, China. Atmosphere 12(10):1271

Zhang Y, Murray AT, TurnerIi BL (2017b) Optimizing green space locations to reduce daytime and nighttime urban heat island effects in Phoenix, Arizona. Landsc Urban Plann 165:162–171

Zhao H, Tan J, Ren Z, Wang Z (2020) Spatiotemporal characteristics of urban surface temperature and its relationship with landscape metrics and vegetation cover in rapid urbanization region. Complexity. https://doi.org/10.1155/2020/7892362

Zhao L, Oppenheimer M, Zhu Q, Baldwin JW, Ebi KL, Bou-Zeid E et al (2018) Interactions between urban heat islands and heat waves. Environ Res Lett 13(3):034003. https://doi.org/10.1088/1748-9326/aa9f73

Zhao Y, Zhong L, Ma Y, Fu Y, Chen M, Ma W, Zhou K (2021) WRF/UCM simulations of the impacts of urban expansion and future climate change on atmospheric thermal environment in a Chinese megacity. Clim Change 169(3):38

Zhou W, Huang G, Cadenasso ML (2011) Does spatial configuration matter? Understanding the effects of land cover pattern on land surface temperature in urban landscapes. Landsc Urban Plann 102(1):54–63

Zhou Y, Zhao H, Mao S, Zhang G, Jin Y, Luo Y et al (2022) Exploring surface urban heat island (SUHI) intensity and its implications based on urban 3D neighborhood metrics: an investigation of 57 Chinese cities. Sci Total Environ 847:157662. https://doi.org/10.1016/j.scitotenv.2022.157662

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